MCO-1D

Wave Guide

Kinematic hypotheses, learning systems, and ALC reference

v25 · September 2026

The Wave Guide documents the kinematic hypothesis library and its learning infrastructure. v25 adds the ALC (Adaptive Learning Coordinator), three ALC modalities, the regenerative energy recovery system, the flight envelope characterisation framework, and the forward-flight SAR family.

1. The Learning Architecture

Five subsystems, one coordinator.

System	Function	Role
AI Observer	`ai_observer_tick()`	What is happening right now
Discovery	`discovery_hypothesis_select()`	What to try next (ALC-directed)
ALSS	`_ALSS_SEGMENTS`	Did that work? (evaluation scheduling)
Genome	`_MW_* (15 axes)`	What to remember (weighted mechanisms)
CMTS	`_CMTS_CHAMPIONS (11)`	Does what worked here also work elsewhere?
ALC ★	`_ALC_SHARED`	Coordinates all five, sterile cockpit, pilot interface

★ ALC is a coordinator, not a replacement. All five continue if ALC fails.

2. Hypothesis Families

H-ACC (7 hypotheses)

`H-ACC-1` Stability switch — unstable cruise to stable tuck (Cooper's hawk)

`H-ACC-2` Gap stroke phase — upstroke hold at minimum span

`H-ACC-3` Sideways aperture passage — body roll + asymmetric sweep

`H-ACC-4` Ballistic gap tuck — body coasts through, wings stopped neutral

... and 3 more

H-AM (4 hypotheses)

`H-AM-1` Shuttlecock brake — max pressure drag at 90° AoA

`H-AM-2` Lift dump — neutral AoA spoiler effect for approach

`H-AM-3` J-hook post-stall perch — forward sweep + 85° AoA

`H-AM-4` Lag displacement roll — 90° phase offset roll authority

H-AQ (3 hypotheses)

`H-AQ-1` Whisper cruise — low frequency, high amplitude, Strouhal-neutral

`H-AQ-2` Minimum tip velocity at Vne — acoustic sprint mode

`H-AQ-3` Whisper pigeon — reduced-frequency dihedral descent: quieter than pige

H-BS (3 hypotheses)

H-BS-1 Snow surface adaptive takeoff — weight-forward to break crust, then pi

H-BS-4 Crosswind ice operations — wing amplitude differential provides direct

H-BS-5 Nap-of-earth terrain following — constant AGL altitude over undulating

H-CE (3 hypotheses)

H-CE-1 High-altitude amplitude scaling — compensate for thin air

H-CE-2 Low-Re wake capture optimisation — unsteady advantage

H-CE-3 Ceiling hold stability — wide dihedral + slow flap in thin air

H-CLB (4 hypotheses)

H-CLB-1 Best rate of climb (V_y) — maximum Ps at 9-10 m/s

H-CLB-2 Best angle of climb (V_x) — maximum (T-D) at 6-8 m/s

H-CLB-3 Zoom climb — kinetic-to-potential energy conversion

H-CLB-4 Sustained climb — best ROC within rope and motor thermal limits

H-CR (5 hypotheses)

H-CR-1 GPS-denied hover stability — cascade tuning for minimum drift when pos

H-CR-2 Payload release kinematics — cascade profile minimises attitude distur

H-CR-3 Cargo cruise efficiency — minimum power per kg·km with payload attache

H-CR-4 High-altitude surveillance loiter — maximum time-on-station at altitud

... and 1 more

H-DA (7 hypotheses)

H-DA-1 Density-adaptive hover — real-time ρ estimation from pitot + IMU

H-DA-2 Amplitude-first density compensation — bumblebee Everest strategy

H-DA-3 Thin-air unsteady advantage — LEV and wake capture optimisation at low

H-DA-4 High-density-altitude clap intensification — clap-and-fling as primary

... and 3 more

H-DIH (6 hypotheses)

H-DIH-1 Geometric dihedral sweep — find optimal restoring angle for passive ro

H-DIH-2 Stroke-dihedral asymmetry — upstroke higher than downstroke for passiv

H-DIH-3 Weathercock cascade — cascade phase profile that mimics vertical tail

H-DIH-4 Coordinated turn geometry — proverse yaw amplitude ratio tuning for ru

... and 2 more

H-EN (3 hypotheses)

H-EN-1 Sprint to hover — minimum stopping distance from max speed, cascade re

H-EN-2 Stoop to lateral translation — convert dive kinetic energy into sidewa

H-EN-3 Direction reversal — rearward flight to forward sprint without stoppin

H-EV (2 hypotheses)

H-EV-1 MIL-STD-1797B handling qualities — bandwidth, phase delay, quickness i

H-EV-2 Acoustic certification — eVTOL 65 dB target at 76m using Gutin tonal m

H-FD (5 hypotheses)

H-FD-1 Pigeon steep descent — high-dihedral stroke plane controls glide path

H-FD-2 Dihedral glidepath — dihedral angle sets descent rate independently of

H-FD-3 Differential dihedral steering — raise one wing to steer descent preci

H-FD-4 Near-vertical dihedral brake — 70-85° wing angle, maximum aerodynamic

... and 1 more

H-FG (3 hypotheses)

H-FG-1 Conservative flap-glide (3s flap / 2s glide)

H-FG-2 Aggressive flap-glide (2s flap / 3s glide)

H-FG-3 Dynamic flap-glide (optimizer-controlled ratio)

H-GS (1 hypotheses)

H-GS-1 Per-wing gust suppression — individual wing current monitors pre-adapt

H-HB (5 hypotheses)

H-HB-1 Hummingbird stroke-plane turn — cascade tilt vectors aerodynamic force

H-HB-2 Inner/outer wing asymmetry — outer wing higher and flatter, inner wing

H-HB-3 Drag-based yaw precision — LEV strength asymmetry controls turn rate w

H-HB-4 Hummingbird escape manoeuvre — maximum-authority 180° yaw reversal usi

... and 1 more

H-HL (5 hypotheses)

H-HL-1 Maximum hover thrust — all 8 wings at peak, clap armed, cascade optimi

H-HL-2 Payload-adaptive trim — cascade and dihedral re-optimise automatically

H-HL-3 Cargo pendulum suppression — cascade differential damps suspended load

H-HL-4 Ballistic zoom to altitude — convert cruise kinetic energy to maximum

... and 1 more

H-HS (4 hypotheses)

H-HS-1 Drag-minimised maximum sweep — high-speed drag reduction above 18 m/s

H-HS-2 Burst-coast sprint — timed flap bursts at 18–24 m/s

H-HS-3 Fixed-wing cruise mode — rear pairs lift, front pair thrust only

H-HS-4 Strouhal-optimal transition — 12–16 m/s transition zone tuning

H-KW (4 hypotheses)

H-KW-1 Kestrel body angle equilibrium — stopped wings, pitched body

H-KW-2 Kestrel high-sweep stability — wing sweep as primary control

H-KW-3 Kestrel coupled dihedral-sweep synergy — Royal Society Interface 2026

H-KW-4 Orographic hover over obstacle — updraft core tracking

H-LB (3 hypotheses)

H-LB-1 Level-body braking — front/rear cascade differential

H-LB-2 Level-body acceleration — front thrust + rear lift bias

H-LB-3 Emergency level-body deceleration — maximum cascade reversal

H-MO (3 hypotheses)

H-MO-1 Maximum sweep at Vne approach — drag minimisation

H-MO-2 Front-pair frequency boost — rear-pair LEV surfing

H-MO-3 Ground effect hover — frequency reduction near surface

H-MV (4 hypotheses)

H-MV-1 Proverse yaw — inside-wing amplitude boost in turns

H-MV-2 Rapid deceleration — collective high-AoA stall brake

H-MV-3 Phase-advance pitch authority — faster nose response

H-MV-4 Split-cascade maneuver authority — per-pair independent phase

H-OV (3 hypotheses)

H-OV-1 80 Jump — cascade angle that maximises simultaneous altitude and airsp

H-OV-2 Pure lateral translation — sideward hover flight without yaw, roll sti

H-OV-3 Combined rearward+lateral translation — diagonal, arced lateral, and f

H-PDW (3 hypotheses)

H-PDW-1 Stability triangle — proverse / dihedral / weathercock joint optimisat

H-PDW-2 Speed-adaptive proverse-dihedral ratio — coordination balance changes

H-PDW-3 Dutch roll suppression — proverse yaw and weathercock cascade tuned to

H-PERF (7 hypotheses)

H-PERF-1 Top speed trial — maximum level-flight airspeed at full lever, best wi

H-PERF-2 Cruise efficiency trial — minimum Wh/km at target airspeed in cruise z

H-PERF-3 Hover efficiency trial — maximum N/W (figure of merit) in trim detent

H-PERF-4 Intent latency trial — minimum time from stick input to measured aircr

... and 3 more

H-PF (6 hypotheses)

H-PF-1 Fixed-wing pitch feel — aft stick produces proportional VSI response w

H-PF-2 Back-side slow flight — power controls altitude, pitch controls airspe

H-PF-3 Nose attitude continuity — fuselage pitch rate < 3 deg/s through 3-5 m

H-PF-4 Hover departure nose management — smooth nose attitude evolution from

... and 2 more

H-PH (4 hypotheses)

H-PH-1 In-phase sprint — collapsed metachronal cascade

H-PH-2 LEV capture phase sweet spot (37° offset at 9 m/s)

H-PH-3 LEV capture phase — high-speed variant (22° at 15 m/s)

H-PH-4 Anti-phase loiter — minimum power, maximum endurance

H-PREC (5 hypotheses)

H-PREC-1 Trim-optimised cruise — minimum AoA correction at 50% lever

H-PREC-2 Heading-stable cruise — cascade phase for zero yaw torque

H-PREC-3 Seaplane water approach — VSI ±100 ft/min on final below 50m

H-PREC-4 IAS stability — minimum thrust variation at cruise lever

... and 1 more

H-PULL (1 hypotheses)

H-PULL-1 High-load pitch authority — progressive cascade pullout, knife-edge wi

H-RC (4 hypotheses)

H-RC-1 High-bank turn tightening — aft pitch past 0.7 at bank >45° applies ca

H-RC-2 Rearward flight — aft pitch >0.5 + both pedals, arc steering from peda

H-RC-3 Hard brake — lever 0-3% + pitch neutral: maximum deceleration, body fl

H-RC-4 Nose attitude control — both pedals depressed, pitch stick sets nose a

H-REGEN (2 hypotheses)

H-REGEN-1 Regenerative descent — wing config that recovers maximum energy in des

H-REGEN-2 Regenerative braking — active brake and emergency stop zones return ma

H-RES (3 hypotheses)

H-RES-1 Single wing pair loss — cascade redistribution across 7/8 pairs mainta

H-RES-2 Two wing pair loss — controlled flight maintained with 6/8 pairs, degr

H-RES-3 Fault detection and adaptation latency — time from wing pair failure t

H-SAR (4 hypotheses)

H-SAR-FF-1 Forward-flight SAR orbit — tight slow circle at V_min for sensor dwell

H-SAR-FF-2 Minimum-speed reversal turn — tightest altitude-stable 180° direction

H-SAR-FF-3 Sensor orbit approach — curved descending approach that keeps the targ

H-SAR-FF-4 Forward-flight payload drop — minimum-speed pass over target with prec

H-SE (2 hypotheses)

H-SE-1 Surface effect mode — WIG + low-altitude flap-glide over water

H-SE-2 Surface effect cruise — WIG transit at 1m AGL over water

H-SK (3 hypotheses)

H-SK-1 High-AoA power stroke — maximum mid-stroke pitch

H-SK-2 Port-stbd phase split — clap-and-fling approach

H-SK-3 Asymmetric amplitude — power-stroke bias

H-SOR (6 hypotheses)

H-SOR-1 Glide polar optimisation — minimum sink wing configuration

H-SOR-2 Best glide configuration — maximum L/D for cross-country

H-SOR-3 Slow flap assist — minimum power addition for sustained near-zero sink

H-SOR-4 Thermalling configuration — tight circle, minimum span, low wing loadi

... and 2 more

H-SP (2 hypotheses)

H-SP-1 Nose-down sensor hover — front-pair forward cascade for downward point

H-SP-2 Nose-up sensor hover — rear-pair cascade for upward pointing

H-ST (3 hypotheses)

H-ST-1 Strouhal-optimal cruise frequency (9 m/s)

H-ST-2 Amplitude-speed trade at 15 m/s (constant St)

H-ST-3 Reduced-frequency loiter (St=0.25, 6 m/s)

H-STOOP (1 hypotheses)

H-STOOP-1 Dive dart — speed-adaptive wing area maximises dive speed within struc

H-STORM (7 hypotheses)

H-STORM-1 Gust-damping cascade phase (tandem-wing passive attenuation)

H-STORM-2 Asymmetric amplitude fast gust compensation

H-STORM-3 Maximum dihedral stability configuration

H-STORM-4 Storm headwind penetration — tilted stroke plane

... and 3 more

H-TR (3 hypotheses)

H-TR-1 Speed-adaptive pair handoff — progressive thrust/lift transition

H-TR-2 Transition hysteresis — preventing oscillation at crossover speed

H-TR-3 Inside-pair cascade reversal for turn radius compression

H-TX (1 hypotheses)

H-TX-1 Seamless hover-to-cruise transition — cascade blend rate optimisation

H-VIFF (3 hypotheses)

H-VIFF-1 Differential VIFF — outside-wing AoA increase for centripetal force

H-VIFF-2 Symmetric VIFF — collective AoA increase, instantaneous turn rate

H-VIFF-3 Thrust-vector yaw — nose-pointing without banking

H-WC (2 hypotheses)

H-WC-1 Dorsal clap-and-fling at maximum validated amplitude

H-WC-2 Anti-phase maximum amplitude — extended stroke in pure cruise

H-WD (3 hypotheses)

H-WD-1 Water departure clap auto-arm — seagull 22% lift synergy between clap-

H-WD-2 Water departure phase coordination — front pairs thrust, rear pairs li

H-WD-3 Water departure asymmetric stroke — slow upstroke, fast downstroke Q=2

3. New Hypotheses — This Session

H-OV-3 Combined rearward+lateral translation — diagonal, arced lateral, and full-e

SYNTHESIS OF H-RC-2 AND H-OV-2: H-RC-2 (rearward flight) uses cascade reversal (90°-135°) to drive the aircraft backward. H-OV-2 (lateral translation) uses amplitude differential to drive the aircraft

H-EN-1 Sprint to hover — minimum stopping distance from max speed, cascade reversa

BACKGROUND KINEMATICS: Pilot commands stop (lever to brake zone). Genome optimises the cascade reversal profile — the specific sequence of cascade angles and amplitudes as the aircraft decelerates fro

H-EN-2 Stoop to lateral translation — convert dive kinetic energy into sideward ve

BACKGROUND KINEMATICS: The aircraft has completed a stoop and has significant kinetic energy. Instead of a conventional pullout (H-PULL-1 Mode A), the pilot deflects the roll stick at the bottom to re

H-EN-3 Direction reversal — rearward flight to forward sprint without stopping, ca

THE CHEYENNE MANOEUVRE: The AH-56 Cheyenne demonstrated the ability to fly backward and then transition immediately to forward flight without stopping. The prop pitch reversal made this possible — the

H-RES-1 Single wing pair loss — cascade redistribution across 7/8 pairs maintains c

BACKGROUND KINEMATICS — SAFETY CRITICAL: When one wing pair fails (ODrive fault, servo jam, broken winglet), the remaining 7 pairs must be reconfigured to: (a) Maintain lift (total thrust is 87.5% o

H-RES-2 Two wing pair loss — controlled flight maintained with 6/8 pairs, degraded

BACKGROUND KINEMATICS — SAFETY CRITICAL: Loss of two pairs (25% of wing area) is a more severe fault scenario. The aircraft can still fly but is significantly degraded. Two scenarios: (a) Both lost

H-RES-3 Fault detection and adaptation latency — time from wing pair failure to sta

BACKGROUND KINEMATICS — SAFETY CRITICAL: The speed of fault detection determines how much attitude deviation the aircraft experiences before adaptation begins. At 50 Hz sensing, the minimum detection

H-AQ-3 Whisper pigeon — reduced-frequency dihedral descent: quieter than pigeon OR

SYNTHESIS OF H-AQ-1 (WHISPER) AND H-FD-3/4 (PIGEON STEEP): The insight: dihedral braking (H-FD-3/4) generates descent through aerodynamic drag from high-dihedral wings, NOT through cascade reversal.

H-FD-5 Pigeon descent with lateral repositioning — descend flat-body while transla

SYNTHESIS OF H-FD-3 (PIGEON DIFFERENTIAL) AND H-OV-2 (LATERAL): H-FD-3 uses differential dihedral to steer the descent path — raising one wing descends toward the other side. H-OV-2 uses amplitude di

H-HL-5 Heavy lift lateral repositioning — translate sideways with suspended cargo,

SYNTHESIS OF H-HL-3 (PENDULUM SUPPRESSION) AND H-OV-2 (LATERAL): H-HL-3 suppresses fore-aft pendulum oscillation during hover with a suspended load. H-OV-2 translates laterally in hover. H-HL-5 addre

H-REGEN-1 Regenerative descent — wing config that recovers maximum energy in descent/

STEWARDSHIP PRINCIPLE: Any energy that would become heat in motor windings during braking or descent should instead return to the battery. The aerodynamic load on the wings during descent can overdriv

H-REGEN-2 Regenerative braking — active brake and emergency stop zones return maximum

STEWARDSHIP PRINCIPLE: The active brake and emergency stop zones apply maximum aerodynamic drag via cascade reversal. That drag force does mechanical work on the motor shafts. Currently that work beco

H-PERF-6 First-flight performance characterisation — map complete flight envelope, m

THE MOST IMPORTANT HYPOTHESIS IN THE SYSTEM: Run on the first flight, before any other hypothesis. Maps the aircraft's actual flight envelope from measured data rather than pre-flight estimates. The

H-PERF-7 Minimum airspeed refinement — find wing configuration that achieves lowest

Once H-PERF-6 establishes the initial V_MIN_MS, H-PERF-7 asks whether a different wing configuration can sustain level flight at a lower airspeed. This is important because V_MIN_MS directly sets the

H-PREC-2 Heading-stable cruise — cascade phase for zero yaw torque

Heading tracking requires the cascade phase to produce zero net yaw torque at cruise. Small cascade asymmetries between the four pairs produce systematic heading drift even in still air. This hypothes

H-SAR-FF-1 Forward-flight SAR orbit — tight slow circle at V_min for sensor dwell over

FORWARD-FLIGHT SAR PRIMARY MODE: When HOVER_CAPABLE = False (or as an endurance-efficient alternative when hover is available), the aircraft maintains continuous observation of a point target by flyin

H-SAR-FF-2 Minimum-speed reversal turn — tightest altitude-stable 180° direction rever

BACKGROUND KINEMATICS — PILOT-INVISIBLE: When the pilot reverses direction at low speed — whether at the end of a search pass, repositioning between targets, or manoeuvring in confined airspace — the

H-SAR-FF-3 Sensor orbit approach — curved descending approach that keeps the target co

FORWARD-FLIGHT SAR APPROACH MODE: In hover-capable aircraft, the pilot descends vertically to the target. Without hover, the approach must be curved — a descending spiral that keeps the target in the

H-SAR-FF-4 Forward-flight payload drop — minimum-speed pass over target with precise p

FORWARD-FLIGHT SAR PAYLOAD MODE: Without hover, payload delivery requires a low-speed pass directly over the target. The payload is released at the moment of minimum airspeed (closest to a drop from h

4. Autopilot Deferred (12 items)

The autopilot boundary: background kinematics decide how wings move. Autopilot decides where the aircraft goes. Items below require path planning or trajectory management.

`H-SAR-1` — Expanding square search pattern — path planning required

`H-SAR-2` — Terrain contour search — path planning required

`H-SAR-3` — Confined space approach — combined autopilot + sensor fusion

`H-SAR-4` — Victim vector approach — path planning from sensor fix

`H-SOR-7` — Thermal exit sprint — thermal exploitation autopilot decision

`RTL` — Return to launch — autopilot post first flight

`POSITION_HOLD` — GPS/vision position hold — autopilot post first flight

`COURSE_HOLD` — Course hold with wind correction — autopilot

`PATTERN_FLIGHT` — Autonomous search patterns — autopilot

`ORBIT_AUTONOMOUS` — Autonomous SAR orbit — software chooses orbit centre, radius, and entry/exit; pi

`LAWNMOWER_PATTERN` — Autonomous lawnmower search — software chooses track spacing, flies parallel tra

`DESCENDING_APPROACH_AUTO` — Autonomous descending approach to target — software guides aircraft along comput

5. Key Cross-Talk Links (built this session)

Link	Description
Observer → Discovery	When observer detects anomaly, relevant hypothesis immediately enters ALC priority queue. Discovery selector checks queue first.
CMTS → Discovery	When cross-mode winner found, burst evaluation queued for all related contexts within next 10 evaluation cycles.
Genome → Observer	When genome axis top mechanism changes, shift logged for post-flight debrief. Pilot learns how genome is evolving.
Session memory	Hypothesis evaluation history, parameter region exhaustion, and improvement velocity tracked across all flights.
Improvement velocity	8-evaluation rolling window per genome axis. When velocity drops below threshold, discovery search radius widens.
Degradation detection	5% metric degradation in 2 consecutive evaluations triggers Tier 2 or 3 alert and queues relevant hypotheses for investigation.